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AI Mobile Engineer

Your AI mobile app might work perfectly, but exposing your AI API key can put your entire application at risk.





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Why Do You Need an AI Backend?

- AI Backend: A server-side layer that securely connects your mobile app with AI services.
- Secure Communication: Keeps sensitive operations away from the mobile client.
- Centralized Control: Manages AI requests, users, usage, and application logic.
- Production Ready: Makes AI applications easier to secure, monitor, and scale.





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Never Call LLMs Directly from the Client

- Mobile Client: Runs on the user's device and can be inspected or reverse-engineered.
- Exposed API Keys: Keys embedded in mobile apps can be extracted and misused.
- Unauthorized Usage: Attackers may use your API key and generate unexpected costs.
- Solution: Send AI requests through your secure backend instead.





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Proxy Pattern

- Proxy Pattern: Your backend acts as an intermediary between the mobile app and the AI provider.
- Request Flow: Mobile App → AI Backend → LLM Provider.
- Response Flow: LLM Provider → AI Backend → Mobile App.
- Backend Responsibilities: Authenticate users, validate requests, attach API keys, and process responses.





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Secrets Management

- Secrets: Sensitive credentials such as AI API keys and database passwords.
- Environment Variables: Store secrets outside your application source code.
- Backend Only: Never expose private API keys in mobile app code or public repositories.
- Production: Use secure environment configuration or a dedicated secrets manager.





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Rate Limiting

- Rate Limiting: Controls how many requests a user can make within a specific time.
- Cost Protection: Prevents excessive AI API usage and unexpected expenses.
- Abuse Prevention: Helps protect your backend from spam and automated requests.
- Example: Allow a maximum of 10 AI requests per user per minute.
- Additional Protection: Combine rate limiting with authentication, usage quotas, and monitoring.

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Why NestJS?

- NestJS: A TypeScript-based Node.js framework for building scalable server-side applications.
- Modular Architecture: Organizes backend code into modules, controllers, and services.
- Dependency Injection: Helps manage and reuse application services.
- API Development: Supports REST APIs, streaming, authentication, and integrations.
- Our Choice: We'll use NestJS to build the AI backend for our mobile applications.



